INCH-POUND

MIL-T-46598F (AR)
30 June 1992
SUPERSEDING
MIL-T-46598E (MU)
29 September 1976

MILITARY SPECIFICATION

TRACER, PROJECTILE: M13

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies for the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the M13 Tracer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection, and Testing.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, Pennsylvania 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.8)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

8860550 Tracer, Projectile, M13 Assembly.
8836133 Carton, Packing, Ammunition, for Tracer, XM9 and M13.
8836134 Box, Packing, Ammunition, for Tracer, XM9 and M13.
8796522 Marking Diagram and Sealing for Wooden Packing Boxes.

(Copies of Government documents, drawings and publications, required by suppliers in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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ASTM Methods - AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B117 - Standard Test Method of Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings in 2.1 and the applicable drawings and specifications referenced thereon.

3.2 Components and assemblies. Components and assemblies shall be in accordance with drawings in 2.1 and the applicable drawings and specifications referenced thereon.

3.3 Ballistic test. The tracer shall show visible evidence of burning for 2.0 seconds minimum.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection.

3.5 Workmanship. All parts shall be fabricated and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, flash, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be legible.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation. The first article sample shall consist of the following items in sample quantities as indicated:

55	Tracer Assemblies, complete, including protective coating
5	Cups, including protective coating
5	Cups, uncoated
5	Tubes

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-T-46598F (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Tracer, Projectile: M13 Assembly		See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Tracer Assembly (Drawing 8860550)			
	Examination for defects	55	3.2	4.4.2.6 and 4.4.2.7
	Ballistic tests			
	Hot temperature (+125°F -5°F)	15	3.3	4.5.1
	Ambient temperature (+70°F ±2.5°F)	25	3.3	4.5.1
	Cold temperature (-40°F +5°F)	15	3.3	4.5.1
	Tube (Drawing 8860720)	5	3.2	4.4.2.8
	Cup, Tracer (Drawing 8822518)			
	Examination for defects	5	3.2	4.4.2.1 and 4.4.2.2

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. Lots shall comply with the lot formation provisions of MIL-A-48078. The maximum lot size shall be 150,000 tracers.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

Table II. Attributes sampling inspection

Lot Size	Inspection Level	
	I	II
2 to 150	32	13
151 to 500	32	30
501 to 1200	74	35
1201 to 3200	74	43
3201 to 10000	87	50
10001 to 35000	109	61
35001 to 150000	124	74

Numbers under inspection levels indicate sample size.
If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for both inspection levels.

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b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup, Tracer (prior to applying compound)	1 OF 1	8822518
			NEXT HIGHER ASSEMBLY: 8860550

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter, min.	Level I	3.2	Gage
102	Length, min.	Level I	3.2	Gage
103	Protective finish improper	1/	3.2	ASTM Method B-117 (salt spray test)
Minor				
201	Inside diameter	Level II	3.2	Gage
202	Thickness thru bottom	Level II	3.2	Gage
203	Burrs	Level II	3.5	Visual
204	Foreign matter	Level II	3.5	Visual
205	Crack, puncture or other damage	Level II	3.5	Visual
206	Workmanship	Level II	3.5	Visual

NOTES: 1/ Thirty randomly selected samples. A failure shall cause the lot to be rejected.

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CLASSIFICATION OF CHARACTERISTICS

MIL-T-46598F (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cup, Tracer (prior to insertion of tube)	1 OF 1	8822518
			NEXT HIGHER ASSEMBLY 8860550

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Coating compound missing, or not completely covering interior surface of cup	Level I	3.2	Visual
Minor 201	Workmanship	Level II	3.5	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Carton, Packing, (prior to sealing)	1 OF 1	8836133
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Any packing components missing	Level II	3.2	Visual
102	Tracer missing	Level II	3.2	Visual
Minor				
201	Workmanship	Level II	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-T-46598F (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Carton, Packing, Sealing	1 OF 1	8836133
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Sealing strip fails to completely seal carton	Level II	3.2	Visual
102	Carton torn, cut or punctured	Level II	3.2	Visual
103	Marking missing, misleading or unidentifiable	Level II	3.2	Visual
Minor				
201	Workmanship	Level II	3.5	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Box, Packing	1 OF 1	8836134
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Box damaged exposing contents	Level I	3.2	Visual
102	Board loose or protruding nail	Level I	3.2	Visual
103	Strapping broken, missing or loose	Level I	3.2	Visual
104	Marking missing, misleading or unidentifiable	Level I	3.2	Visual
105	Cover improperly assembled	Level I	3.2	Visual
106	Contents loose	Level I	3.2	Manual
107	Strapping improperly engaged	Level I	3.2	Visual
Minor				
201	Workmanship	Level II	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-T-46598F (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Tracer Projectile: M13 Assembly (prior to loading)	1 OF 1	8860550
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Moisture and volatiles content of igniter pellet 1/		3.2	4.5.2
102	Moisture and volatiles content of tracer pellet 1/		3.2	4.5.2
103	Tube missing	Level I	3.2	Visual
Minor				
201	Workmanship	Level II	3.5	Visual

NOTES: 1/ One randomly selected sample from each batch or 8 hours production, whichever is the lesser amount. A failure shall cause the parent quantity represented by the sample to be rejected.

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Tracer Projectile: M13 Assembly	1 OF 1	8860550
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter, maximum	Level I	3.2	Gage
102	Total length, maximum	Level I	3.2	Gage
103	Charge missing or less than minimum required depth from shoulder	Level I	3.2	Visual
104	Igniter charge chipped or broken on any one step in excess of 1/3 of the total area of the step, or total aggregate of chipped or broken area of all steps in excess of 1/3 of the total area of the largest step	Level I	3.2	Visual
105	Ballistic tests			
	Hot temperature (+125°F -5°F)	10 1/	3.3	4.5.1
	Ambient temperature (+70°F ±2.5°F)	10 1/	3.3	4.5.1
	Cold temperature (-40°F +5°F)	10 1/	3.3	4.5.1
Minor				
201	Workmanship	Level II	3.5	Visual

NOTES: 1/ No defects allowed.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-T-46598F (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Tube		8860720
			NEXT HIGHER ASSEMBLY 8860550

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Inside diameter	Level II	3.2	Gage
202	Length	Level II	3.2	Gage
203	Wax missing	Level II	3.2	Visual
204	Workmanship	Level II	3.5	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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4.4.3 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.8 herein.

4.5 Methods of inspection.

4.5.1 Ballistic test. The tracers shall be assembled to and tested in Cartridge, 105mm, TP-T, M490A1. The assembled rounds shall be temperature conditioned for a minimum period of 18 hours and fired from the appropriate weapon. A timing device capable of measuring time to the nearest 0.01 second shall be used to measure burning time. Suitable aids, such as binoculars, should be used if necessary to insure visible contact with the tracer.

4.5.2 Moisture and total volatiles. The moisture and total volatile content of the tracer pellet and igniter pellet shall be determined in accordance with MIL-STD-1234 Method 102.2 (see 6.4).

5. PACKAGING

5.1 Preservation and packaging. The tracer assemblies shall be preserved and packaged in accordance with Drawing C8836133.

5.2 Packing. The cartons shall be packed in accordance with Drawing C8836134.

5.3 Marking. The cartons and boxes shall be marked in accordance with Drawings C8836133, C8836134 and MIL-STD-129.

5.4 Data card. Data card information shall be as specified in MIL-STD-1167.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The item covered by this specification is intended for use in 105mm ammunition.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

6.3 Tracer definition. The tracer as referred to in this specification consists of a tracer pellet and an igniter pellet compressed in a tracer cup.

6.4 Moisture. The igniter and tracer compositions and pellets are hygroscopic and will deteriorate rapidly when exposed to moisture. Small amounts of moisture present in or absorbed by the compositions or pellets, will result in unsatisfactory functioning. The ingredients, compositions and pellets should be safeguarded from moisture and humid atmospheres. It is recommended that the blending and pelleting areas contain an atmosphere of less than 60 percent relative humidity.

6.5 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.6 Hazard notice. The tracer and igniter charge and the tracer and igniter charge composition described herein are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.7 Drawings. Drawings listed in Section 2 of this specification in the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or U.S. Army Armament Research, Development and Engineering Center drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

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6.8 Submission of inspection equipment for design approval. Submit designs as required to: Commander, U.S. Army Armament, Research, Development and Engineering Center, ATTN: SMCAR-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the contract Data Requirement List, DD Form 1423 in the contract.

6.9 Subject term (keyword) listing.
Ammunition
Ballistic Testing
Cartridge
Pyrotechnic

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1315-AC70)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-46598F (AR)	92/06/30

3. DOCUMENT TITLE: TRACER, PROJECTILE: M13

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*)
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: 201-724-6675 (2) AUTOVON: DSN-880-6675
c. ADDRESS (*Include Zip Code*) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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